Angular Advanced - Firebase – Guards

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Goal – Only allow routes to be activated when someone is logged in.

For now – Anonymous authentication

- We can now log in/out of our application.
- But users can enter the `/add` and `/edit` route manually, even if not logged in.
- Prevent this by writing guards
  - We create an `AuthGuard` service
- No need to alter backend/Firebase – all Angular stuff

Create `auth.guard.ts` in folder `shared/guards`

```
import {Injectable} from '@angular/core';
import {CanActivate, Router} from '@angular/router';
import {AuthService} from '../services/auth.service';

import 'rxjs/add/operator/map';

@Injectable()
export class AuthGuard implements CanActivate {

   constructor(private router: Router,
            private authService: AuthService) {
   }

   canActivate() {
      return this.authService.authState
         .map(user => {
            if (!user) {
               // user is not authenticated, navigate aw
               this.router.navigate(['/'])
            }
            // else: return boolean of the user
            return !!user;
         })
   }
}
```

Implement CanActivate Interface

Inject Router and AuthService

New! Property in AuthService that returns current user state

If no user exists, navigate away. Else return boolean value of user

Edit `AuthService` to return authState

```
@Injectable()
export class AuthService {

   …
   // getter for current authState
   get authState(){
      return this.firebaseAuth.authState;
   }

}
```

Options. We are using the “official” `get()` notation here.

Could also be done by setting just a simple property, as

the `user` that already exists on this service.

Edit `app.module.ts` to use the Guard

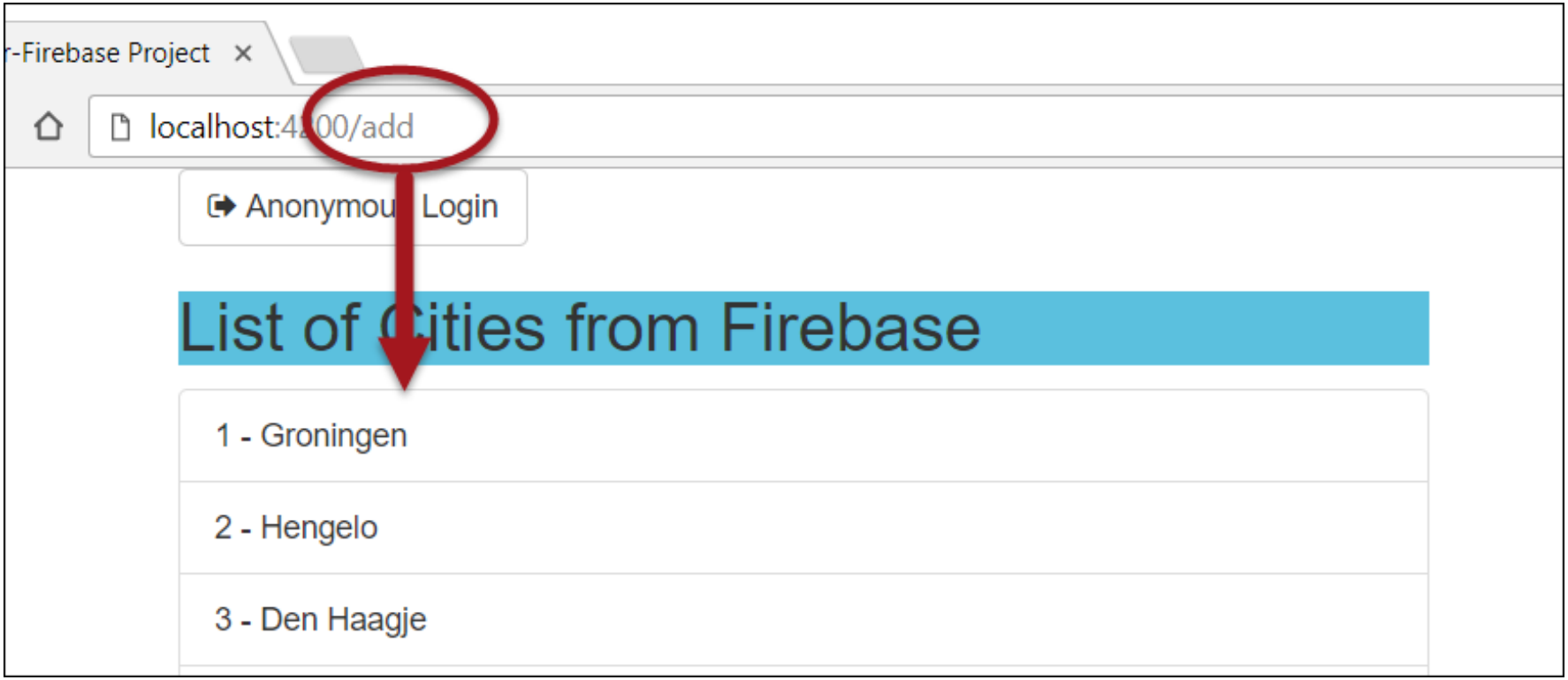
```
// Services
import {AuthGuard} from './shared/guards/auth.guard';


// Initialize Routes
const routes: Routes = [
   {path: '', pathMatch: 'full', redirectTo: 'home'},
   {path: 'home',
      component: HomeComponent}, // always available
   {path: 'add',
      canActivate: [AuthGuard], // guarded
      component: AddComponent},
   {path: 'edit/:key',
      canActivate: [AuthGuard], // guarded
      component: EditComponent},
];

// Initialize Module
@NgModule({
   …
   providers   : [CityService, AuthService, AuthGuard],
   …
})
export class AppModule {}
```

Import Guard

Assign Guard to desired routes

Provide Guard

Test in browser

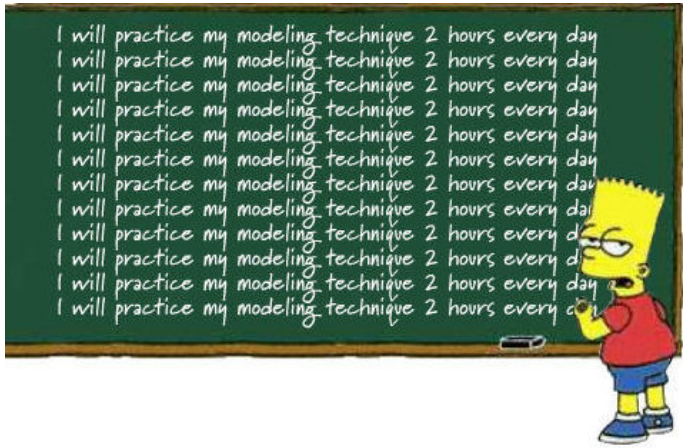


When entering route manually, you should be redirected to the homepage.

Workshop

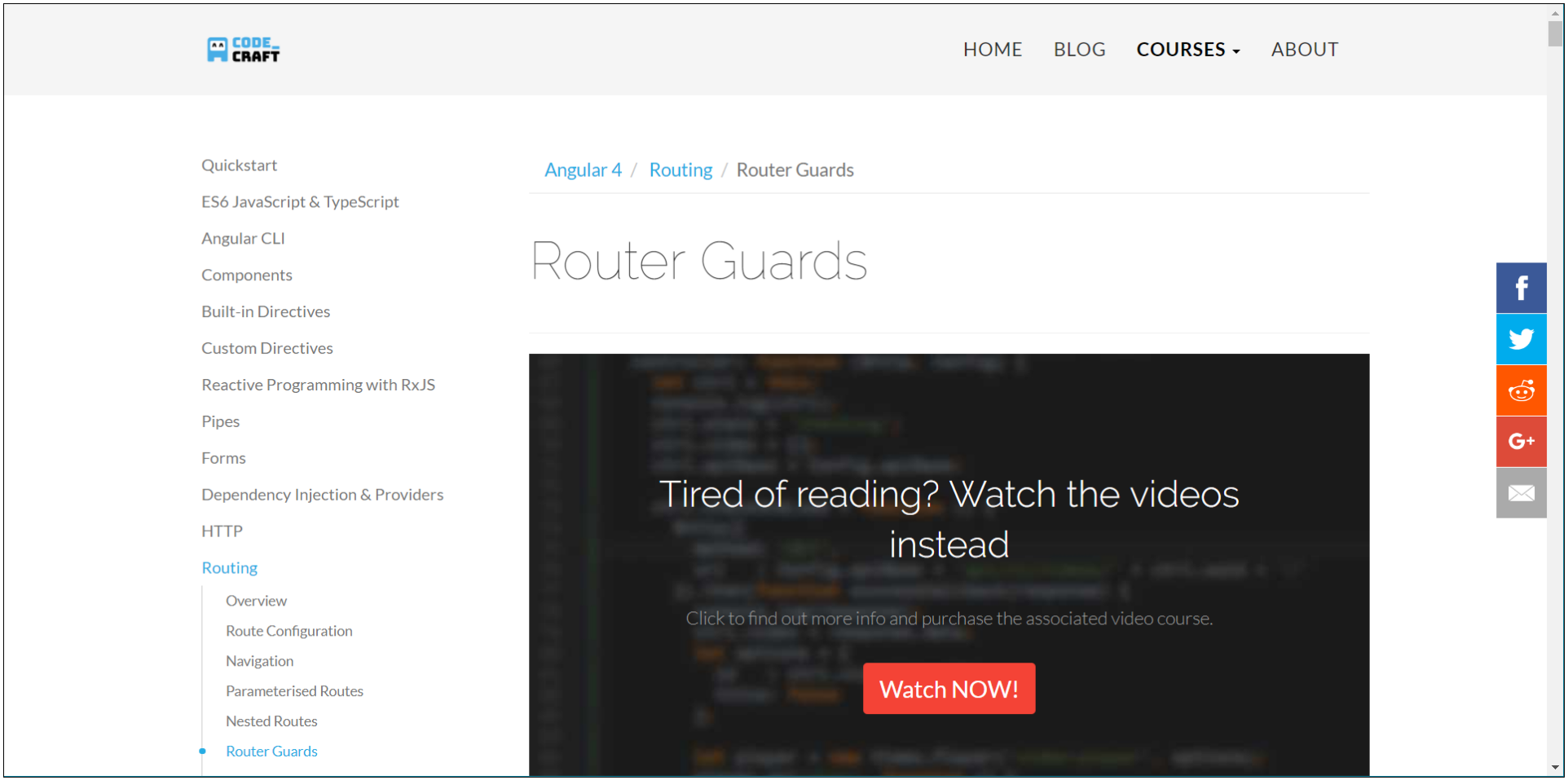
- Add a guard to your application, don't forget to implement `CanActivate()`.
- Use `AuthService` to get current `authState`
- Edit `app.module.ts` to use the guard on guarded routes/components

- Example - `/540-Firebase-guards`



More info

https://blog.thoughtram.io/angular/2016/07/18/guards-in-angular-2.html



https://codecraft.tv/courses/angular/routing/router-guards/